

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics
M2 -PRACTICAL NO - 9

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Class: TYIT	Batch:2
Date of Assignment: 02-09-26	Date/Time of Submission: 02-09-26

9. Hypothesis Testing:

a. Formulate null and alternative hypotheses related to a business problem, conduct a hypothesis test using appropriate statistical tests, and interpret the results.

CODE:-

```
import pandas as pd
from scipy import stats
```

1. Load dataset

```
df = pd.read_csv("supermarket_sales.csv")
```

2. Display first five records

```
print(df.head())
```

3. Select revenue column

```
revenue = df['revenue']
```

4. Check for missing values

```
print("Missing values in revenue:", revenue.isnull().sum())
```

5. Calculate mean revenue

```
mean_revenue = revenue.mean()
print("Mean Revenue:", mean_revenue)
```

6–9. Hypotheses

```
# H0:  $\mu = 1500$ 
# H1:  $\mu \neq 1500$ 
expected_mean = 1500
```

10. Perform one-sample t-test

```
t_stat, p_value = stats.ttest_1samp(revenue, expected_mean)
```

```
print("T-statistic:", t_stat)
print("P-value:", p_value)
```

12–13. Compare p-value with 0.05

```
alpha = 0.05
if p_value < alpha:
    print("Reject H0: Average revenue is significantly different from 1500")
else:
    print("Fail to Reject H0: Average revenue")
```

```
print("KHUSHI T032")
```

OUTPUT:-

```

invoice_id branch      city customer_type gender_customer \
0  750-67-8428      A      Yangon      Member      Female
1  226-31-3081      C  Naypyitaw      Normal      Female
2  631-41-3108      A      Yangon      Normal      Male
3  123-19-1176      A      Yangon      Member      Male
4  373-73-7910      A      Yangon      Normal      Male

product_line unit_cost quantity 5pct_markup revenue \
0  Health and beauty      74.69         7      26.1415  548.9715
1  Electronic accessories      15.28         5       3.8200   80.2200
2  Home and lifestyle      46.33         7      16.2155  340.5255
3  Health and beauty      58.22         8      23.2880  489.0480
4  Sports and travel      86.31         7      30.2085  634.3785

date  time payment_method  cogs  gm_pct gross_income rating
0  01/05/19 13:08      Ewallet  522.83  4.761905      26.1415      9.1
1  03/08/19 10:29      Cash      76.40  4.761905       3.8200      9.6
2  03/03/19 13:23  Credit card  324.31  4.761905      16.2155      7.4
3  1/27/2019 20:33      Ewallet  465.76  4.761905      23.2880      8.4
4  02/08/19 10:37      Ewallet  604.17  4.761905      30.2085      5.3
Missing values in revenue: 0
Mean Revenue: 322.966749
T-statistic: -151.37567896776733
P-value: 0.0
Reject H0: Average revenue is significantly different from 1500
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```

b. Perform A/B testing on a website or marketing campaign to evaluate the effectiveness of different strategies and make data-driven decisions

CODE:-

```

import pandas as pd
from statsmodels.stats.proportion import proportions_ztest
import matplotlib.pyplot as plt

```

1. Load the A/B testing dataset

```
df = pd.read_csv("ab_testing.csv")
```

2. Display the data

```

print("A/B Testing Data:")
print(df)

```

3. Calculate conversion rate for Group A

```

group_a = df[df["Group"] == "A"].iloc[0]
group_b = df[df["Group"] == "B"].iloc[0]

```

```

conversion_rate_a = group_a["Conversions"] / group_a["Users"]
conversion_rate_b = group_b["Conversions"] / group_b["Users"]

```

```

print("\nConversion Rate - Group A:",
      conversion_rate_a * 100, "%")
# 4. Calculate conversion rate for Group B
print("Conversion Rate - Group B:",
      conversion_rate_b * 100, "%")

# 5. Compare conversion rates
print("\nComparison:")
if conversion_rate_a > conversion_rate_b:
    print("Group A has a higher conversion rate.")
elif conversion_rate_b > conversion_rate_a:
    print("Group B has a higher conversion rate.")
else:
    print("Both groups have the same conversion rate.")

# 6 & 7. Hypotheses
print("\nH0: pA = pB")
print("H1: pA ≠ pB")

# 8. Significance level
alpha = 0.05

# 9. Perform two-proportion test
conversions = [group_a["Conversions"], group_b["Conversions"]]
users = [group_a["Users"], group_b["Users"]]

z_stat, p_value = proportions_ztest(conversions, users)

# 10. Display p-value
print("\nZ-statistic:", z_stat)
print("P-value:", p_value)

# 11 & 12. Compare p-value with 0.05
if p_value < alpha:
    print("Decision: Reject the Null Hypothesis")
else:
    print("Decision: Do Not Reject the Null Hypothesis")

# 13. Create bar chart
groups = ["A", "B"]
conversion_rates = [
    conversion_rate_a * 100,
    conversion_rate_b * 100
]

plt.bar(groups, conversion_rates)
plt.xlabel("Group")
plt.ylabel("Conversion Rate (%)")
plt.title("A/B Testing - Conversion Rate Comparison")

```

```
plt.show()
```

14. Conclusion

```
if p_value < alpha:
```

```
    print("\nConclusion: There is a statistically significant difference")
```

```
    print("between the conversion rates of Group A and Group B.")
```

```
else:
```

```
    print("\nConclusion: There is no statistically significant difference")
```

```
    print("between the conversion rates of Group A and Group B.")
```

```
print("KHUSHI T032")
```

OUTPUT:-

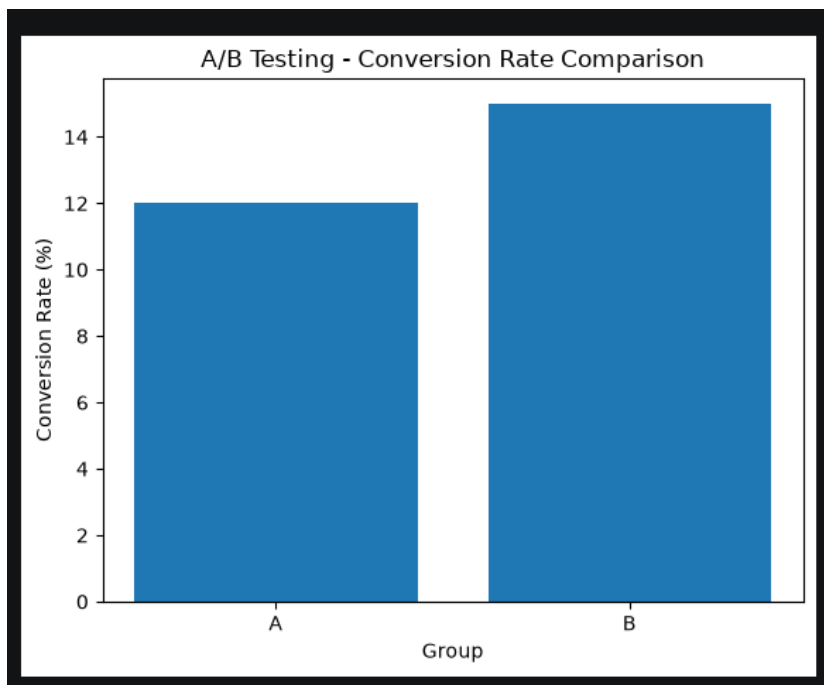
```
A/B Testing Data:
  Group  Users  Conversions
0     A   1000         120
1     B   1000         150

Conversion Rate - Group A: 12.0 %
Conversion Rate - Group B: 15.0 %

Comparison:
Group B has a higher conversion rate.

H0: pA = pB
H1: pA ≠ pB

Z-statistic: -1.963049807622355
P-value: 0.04964038686536878
Decision: Reject the Null Hypothesis
```



```
Conclusion: There is a statistically significant difference
between the conversion rates of Group A and Group B.
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```